Turtle Name Drawer

Nifty Assignment

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How many of us have concerns about the use of AI on students’ assignments and its impact on learning?

When people say we need to think about AI in CS, what does that mean?

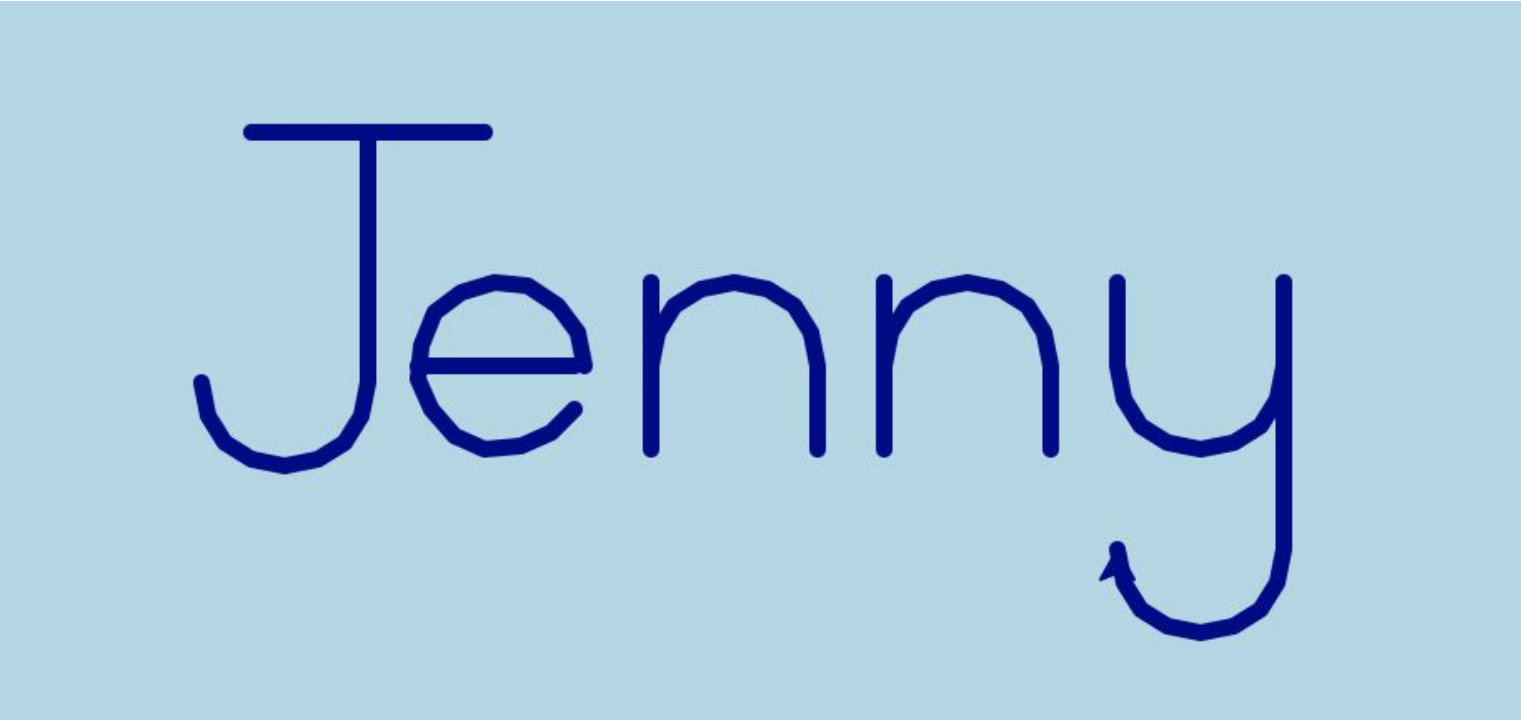
- How to use AI to help us to teach CS?
- How to combate students cheating with AI?
- How to teach students to use AI effectively?

The goal with this assignment is to educate students about the limitations of AI at the beginning of an intro course

- How to use AI to help us to teach CS?
- **How to combate students cheating with AI?**
- **How to teach students to use AI effectively?**

The goal is to go from something like this…

To something like this…



There is little prerequisite knowledge needed

- Python turtle graphics (basic movement methods, creating Turtle and Screen objects, etc.)
- Input() function
- Some guidance on how to use a generative AI tool (e.g., GitHub Copilot) is also helpful

Learning outcomes align with a first assignment

- Debug auto-generated code
- Create an object and assign it to a variable
- Call a method from a given object
- Prompt a user for text input and use their response
- Use comments to help readers understand its functionality

There are few requirements for completion

- Uses python turtles to draw your name
- Accept user input to set some aspect of the drawing
- Correct the given code to draw your name (at least 5 letters) so it looks professional using the turtle module
- Add appropriate comments to the code to explain what is happening

Prompt the AI (we use Github Copilot) to write your name using python turtles

write_jenny_turtle.py

Keep Undo

```python
# Writing "Jenny" with Python Turtles

import turtle

# Set Up the Turtle Environment
screen = turtle.Screen()
screen.bgcolor("white")
screen.title("Write Jenny")

pen = turtle.Turtle()
pen.speed(3)

# Define Functions for Each Letter

def draw_J(pen):
    pen.penup()
    pen.goto(-200, 100)
    pen.pendown()
    pen.setheading(270)
    pen.forward(50)
    pen.circle(25, 180)
    pen.forward(50)

def draw_e(pen):
    pen.penup()
    pen.goto(-150, 100)
    pen.pendown()
    pen.setheading(0)
    pen.circle(25, 180)
    pen.forward(50)
```

CHAT

DRAWING "JENNY" WITH PYTHON T...

write "Jenny" using python turtles

Created write_jenny_turtle.ipynb

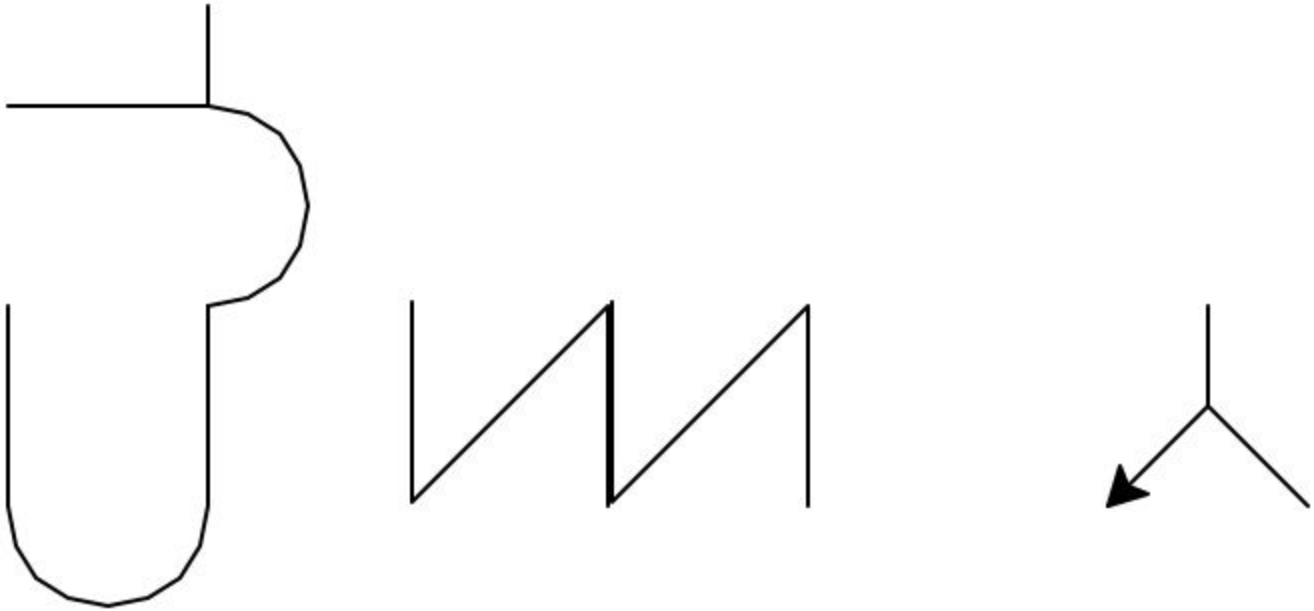
The Jupyter Notebook to write "Jenny" using Python turtles has been created successfully. You can open and run the notebook to see the turtle graphics in action. Let me know if you need further assistance!

GPT-4o • 0x

as a python script

Created

The Python script to write "Jenny" using turtles has been created. You can open and run the script to see the turtle graphics in action. Let me know if you need further assistance!



If you want to avoid the AI using coding features that are too complicated, you can add guardrails

```
# Coding Standards
- No repetition should be used (e.g., while, for, or recursion).
- No functions should be defined.
- No conditionals should be used (e.g., if, elif, else).
- The turtle write method should not be used.
- Use screen.mainloop() to keep the window open.
- Comments should not be overly verbose.
```

EXPLORER

COMP110-STARTER-PSA1

- .github
- .vscode
- .gitignore
- name_drawer.py
- README.md

name_drawer.py

name_drawer.py > ...

```python
"""
Module: name_drawer

A program to draw a name using one (or more) turtles.

Authors:
1) Name - USD Email Address
"""
import turtle

# Set up the screen
screen = turtle.Screen()
screen.title("Drawing 'Jenny'")
screen.bgcolor("white")

# Create the turtle
pen = turtle.Turtle()
pen.speed(1)
pen.pensize(2)

# Draw 'J'
pen.penup()
pen.goto(-200, 100)
pen.pendown()
pen.setheading(270)
pen.forward(100)
pen.circle(50, 180)

# Draw 'e'
```

Keep Undo

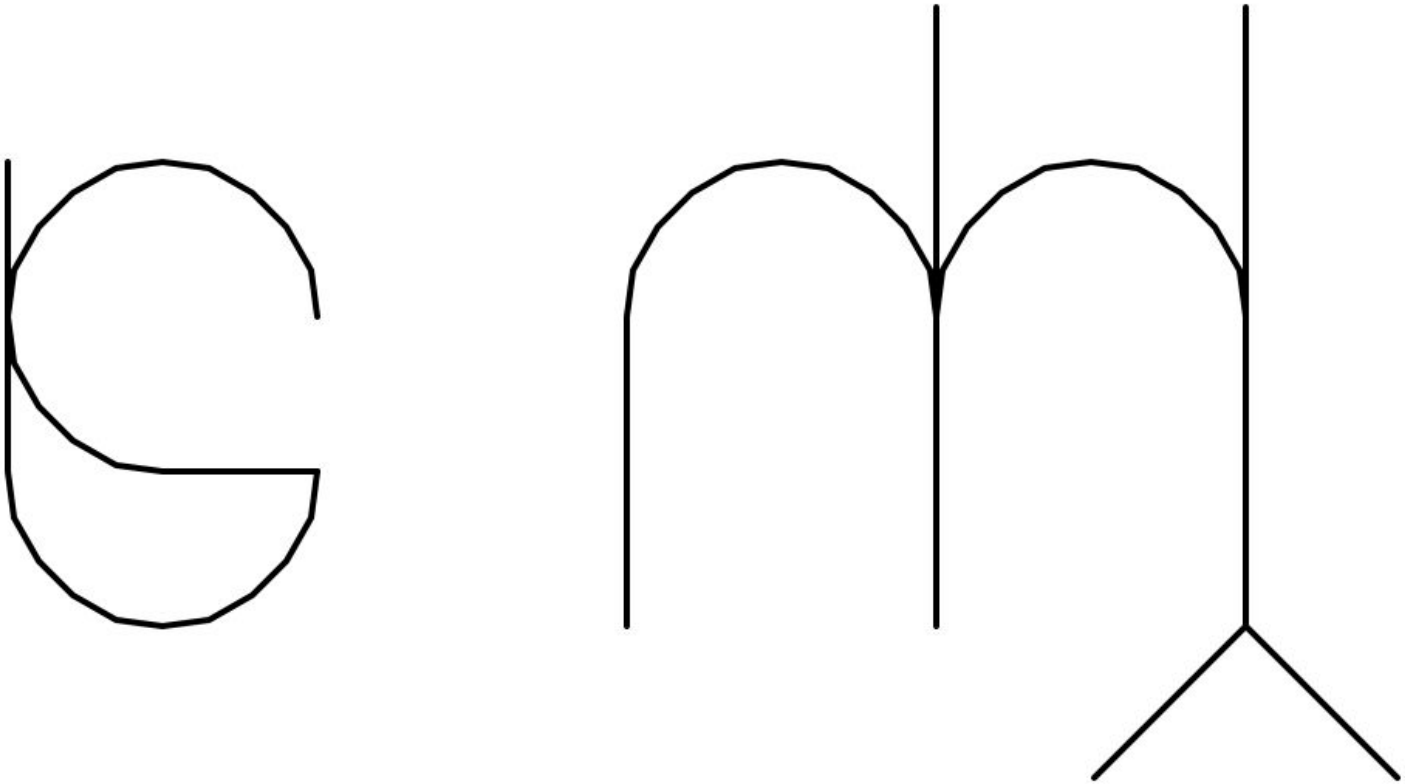
CHAT

DRAWING "JENNY" WITH PYTHON T...

write "Jenny" using python turtles

Edited name_drawer.py +65 -0

I have implemented the Python script to draw the name "Jenny" using turtle graphics. You can run the script to see the output. Let me know if you need further assistance!



Ok, but I saw you were using GPT-4o

EXPLORER

COMP110-STARTER-PSA1

- .github
- .vscode
- .gitignore
- name_drawer.py
- README.md

name_drawer.py

```
5
6   Authors:
    1) Name - USD Email Address
7   1) Jennifer Olsen - USD Email Address
8   """
9   import turtle
10
11  # Set up the screen
12  screen = turtle.Screen()
13  screen.title("Jenny")
14  screen.bgcolor("white")
15  screen.setup(800, 400)
16
17  # Create the turtle
18  pen = turtle.Turtle()
19  pen.speed(3)
20  pen.pensize(3)
21  pen.color("black")
22
23  # --- Draw 'J' ---
24  pen.penup()
25  pen.goto(-250, 100)
26  pen.pendown()
27  # Top bar of J
28  pen.setheading(0)
29  pen.forward(60)
```

Keep Undo

Keep Undo 2 of 2

CHAT

DRAWING "JENNY" WITH PYTHON T...

write "Jenny" using python turtles

Reviewed name_drawer.py

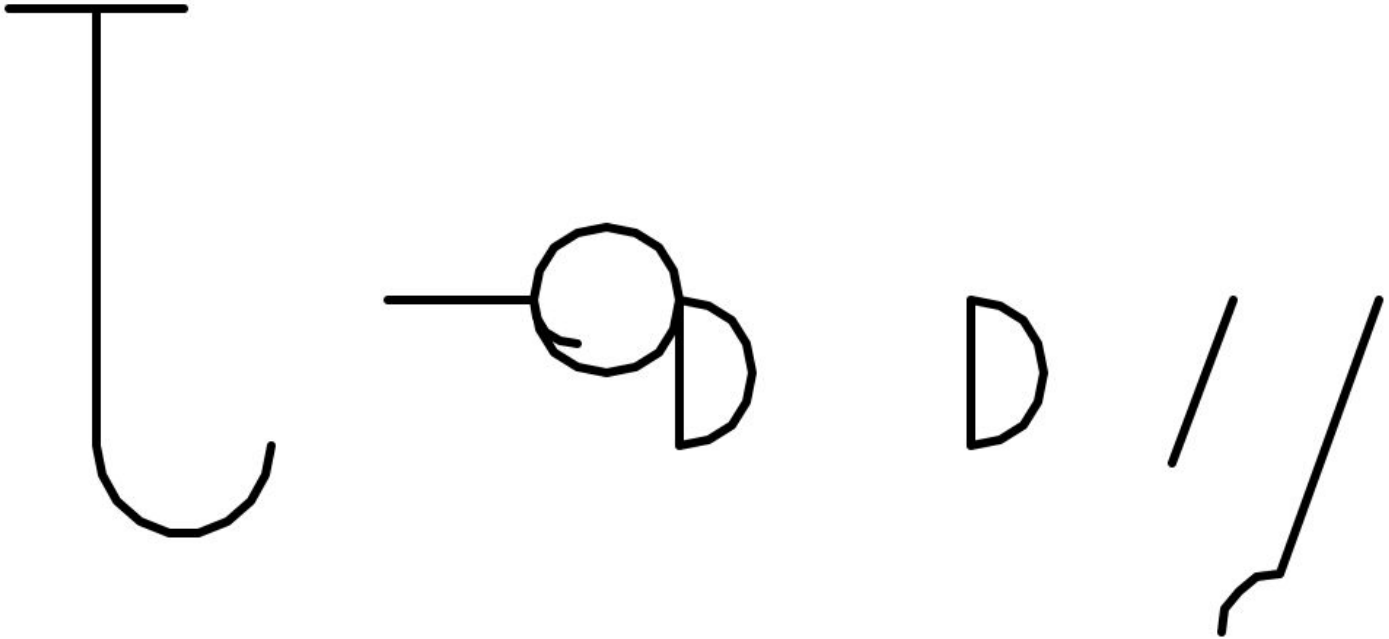
The file was reset to a template. Let me write the full implementation to draw "Jenny" using turtle graphics.

Edited name_drawer.py +99 -1

The script is ready. It draws "Jenny" using turtle graphics with:

- **J** — a top bar, vertical stem, and a curved hook at the bottom
- **e** — a middle bar with top and bottom curves and a small tail
- **n** (×2) — vertical strokes with rounded humps
- **y** — two diagonal strokes with a descending tail and curve

No loops, functions, conditionals, or



No, AI cannot do everything you can do as a CS1 student

You now have to fix the code. There are two options.

1. You can go in and make edits to the code yourself to change it
2. You can prompt the AI about things you want changed and how to make those changes

When looking through the code, you see worked examples

By prompting AI, you get to practice breaking down the problem and defining what you want to happen

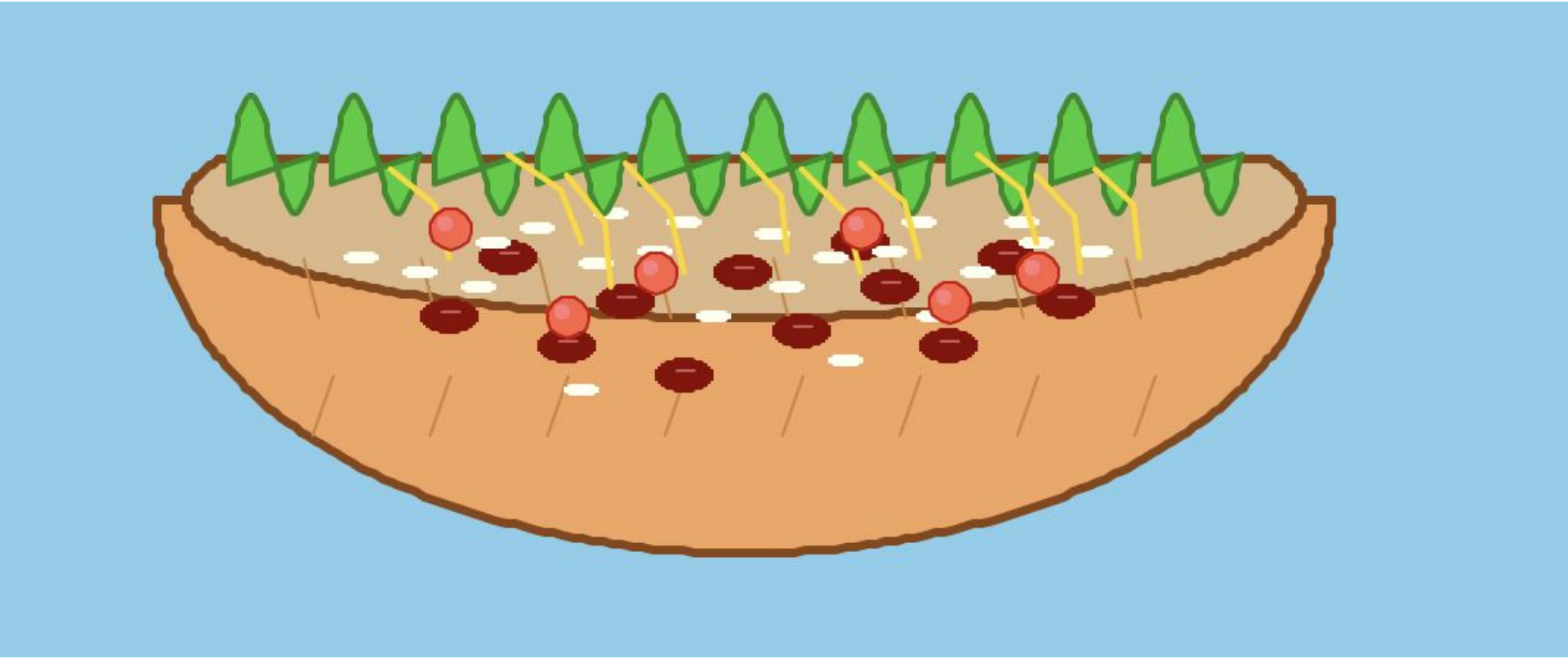
Allowing the use of AI hopefully combats against AI

- It shows students that it is hard to get working code if you do not understand anything
- The AI is not perfect as coding
- Breaking down the problem helps you to use it effectively

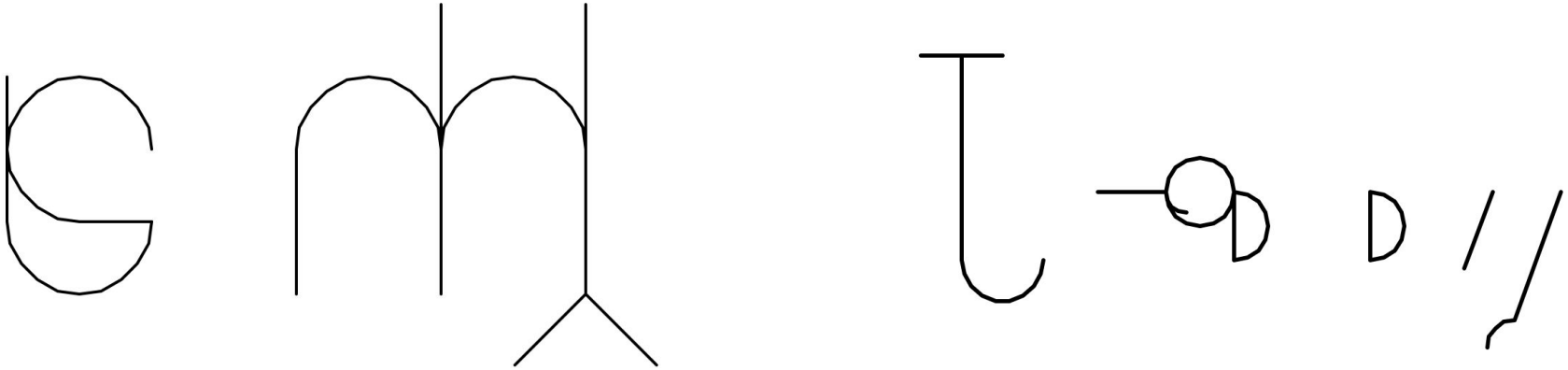
But AI is getting better, soon students will not see that it is not perfect

- We need alternatives for what we can do in the future

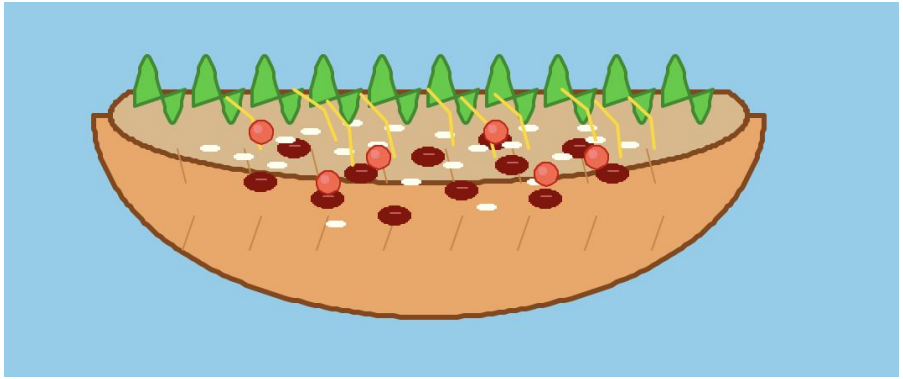
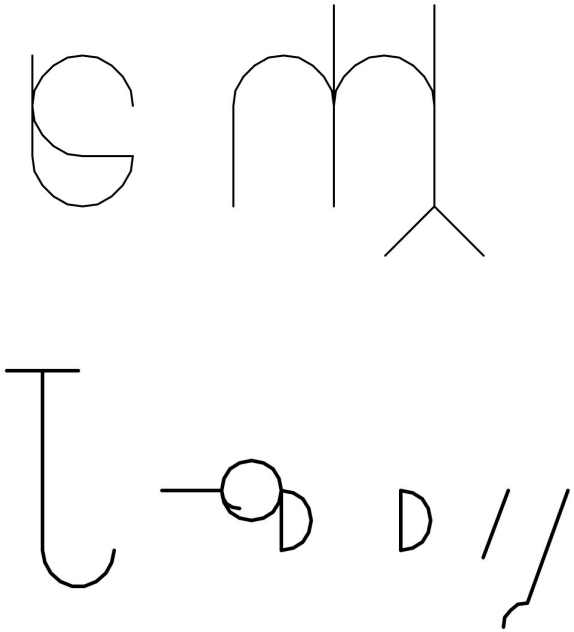
Change the drawing to something not letter based



Compare different models and start a discussion of cost versus accuracy



We all heard how important active learning is, so let’s try it out

If you have questions, please reach out!

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